Merton Model (Stock Analysis)

The Merton model is a mathematical formula that stock analysts and commercial loan officers, among others, can use to judge a corporation's credit default risk. Named for economist Robert C. Merton, who proposed it in 1974, the Merton model assesses the structural credit risk of a company by modelling its equity as a call option on its assets.

Formula for the Merton Model

$$E = V_t N(d_1) - K e^{-r\Delta T} N(d_2)$$

where:

$$d_1 = \frac{\ln \frac{V_t}{K} + \left(r + \frac{\sigma_v^2}{2}\right)\Delta T}{\sigma_v \sqrt{\Delta T}}$$

and

$$d_2 = d_1 - \sigma_v \sqrt{\Delta t}$$

E = Theoretical value of a company's equity

V_t = Value of the company's assets in period t

K = Value of the company's debt

t = Current time period

T = Future time period

r = Risk-free interest rate

N = Cumulative standard normal distribution

e = Exponential term (*i.e.* 2.7183...)

σ = Standard deviation of stock returns

What Does the Merton Model Tell You?

The Merton model allows for easier valuation of a company and helps analysts determine if it will be able to retain solvency, by analyzing the maturity dates of its debt and its debt totals.

The Merton model calculates the theoretical pricing of European put and call options without considering dividends paid out during the life of the option. The model can, however, be adapted to consider dividends by calculating the ex-dividend date value of underlying stocks.

The Merton model makes the following basic assumptions:

- All options are European options and are exercised only at the time of expiration.
- No dividends are paid out.
- Market movements are unpredictable (efficient markets).
- No commissions are included.
- Underlying stocks' volatility and risk-free rates are constant.
- Returns on underlying stocks are regularly distributed.